Daily Algebra Drill #3

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Remainder & Factor Theorem · AM–GM & algebraic inequalities · Completing the square in two variables · Rational Root Theorem · Modular/divisibility arguments · Parametric families of solutions.

Section A Rapid Recognition

1. $f(x) = x^3 - 3x^2 + kx - 8$ has $(x - 2)$ as a factor. Find k.

Ans: $k = 6$

2. Find the remainder when $p(x) = x^{100} - 1$ is divided by $(x - 1)$.

Ans: 0

3. Factorise: $6x^2 - x - 12$.

Ans: $(2x - 3)(3x + 4)$

4. Simplify: $\dfrac{x^{2n}-1}{x^n-1}$.

Ans: x^n+1

5. Given $x>0$, find the minimum value of $x+\dfrac{9}{x}$.

Ans: 6

6. Factorise over $\mathbb{Z}$: x^4-y^4.

Ans: $(x^2+y^2)(x+y)(x-y)$

7. The roots of $2x^2-6x+3=0$ are α and β. Find $\alpha^2+\beta^2$.

Ans: 6

8. Simplify: $\dfrac{(n+1)! - n!}{(n-1)!}$.

Ans: n^2

9. Show that $(a+b+c)^2 - (a^2+b^2+c^2) = 2(ab+bc+ca)$ and state $ab+bc+ca$ when $a+b+c=4$, $a^2+b^2+c^2=8$.

Ans: $ab+bc+ca=4$

10. Find the value(s) of k such that $x^2 + kx + 16$ is a perfect square.

Ans: $k = \pm 8$

Section B Manipulation Drills

1. Find all rational roots of $2x^3 - 3x^2 - 11x + 6$ and factorise completely.

Ans: $(x-3)(x+2)(2x-1)$; rational roots: $x = 3,\ -2,\ \frac{1}{2}$

2. Simplify: $\dfrac{x^3+1}{x^2-x+1}$.

Ans: $x+1$

3. Given $a + b = p$ and $a^3 + b^3 = q$, express ab in terms of p and q.

Ans: $ab = \dfrac{p^3 - q}{3p}$

4. Simplify: $\dfrac{x^2(x+1) - x(x+1)^2}{x(x+1)}$.

Ans: -1

5. Factorise: $a^3 - a^2b - ab^2 + b^3$.

Ans: $(a-b)^2(a+b)$

6. Use $\displaystyle\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$ to find $\displaystyle\sum_{r=1}^{n}(2r-1)$.

Ans: n^2

7. Solve: $\dfrac{2x-1}{x+3} - \dfrac{x+2}{x-1} = 1.$

Ans: $x = -\frac{1}{5}$

8. Simplify: $\left(\dfrac{1}{a} - \dfrac{1}{b}\right) \div \left(\dfrac{1}{a^2} - \dfrac{1}{b^2}\right).$

Ans: $\frac{ab}{a+b}$

9. Show that $n^3 - n$ is divisible by 6 for all integers n.

Ans: Divisible by 6 for all $n \in \mathbb{Z}$.

10. Roots of $x^2 - x - 1 = 0$ are α, β. Find $\alpha^3 + \beta^3$ and $\alpha^4 + \beta^4$.

Ans: $\alpha^3 + \beta^3 = 4$; $\alpha^4 + \beta^4 = 7$

Section C Substitution & Structure

1. Solve: $9^x - 10 \cdot 3^x + 9 = 0$.

Ans: $x = 0$ and $x = 2$

2. Find the minimum value of $x^2 + y^2$ subject to $x + 2y = 5$.

Ans: 5

3. Prove that $a^2 + b^2 + c^2 \geq ab + bc + ca$ for all real a, b, c.

Ans: Proof: see method.

4. Solve: $\sqrt{x+5} + \sqrt{x-3} = 4$.

Ans: $x = 4$

5. Find the minimum of $x^2 + 4x + y^2 - 6y + 13$, and the (x, y) at which it occurs.

Ans: Minimum is 0, at $(x, y) = (-2, 3)$

6. If $x + \dfrac{1}{x} = 5$, find $\dfrac{x^4 + 1}{x^2}$.

Ans: 23

7. Solve over $\mathbb{R}$: $x^4 + 4x^2 - 5 = 0$.

Ans: $x = \pm 1$

8. Given $a, b > 0$ with $a + b = 1$, find the minimum of $\dfrac{1}{a} + \dfrac{1}{b}$.

Ans: 4

Section D Challenge (TMUA / SMC difficulty)

1. Find all real solutions to $x^4 + 2x^3 - 2x - 1 = 0$.

Ans: $x = 1$ and $x = -1$ ($x = -1$ has multiplicity 3)

2. Prove that $n^2 + n + 41$ is not always prime by finding the smallest counterexample, and explain why it is prime for $n = 0, 1, \ldots, 39$.

Ans: Smallest counterexample: $n = 40$, giving $40^2 + 40 + 41 = 41^2 = 1681 = 41 \times 41$.

3. Let $a, b, c > 0$ with $abc = 1$. Prove that $a + b + c \geq 3$.

Ans: Proof via AM–GM: see method.

4. For positive reals a, b, prove $\dfrac{a}{b} + \dfrac{b}{a} \geq 2$. Hence prove AM–GM: $\dfrac{a+b}{2} \geq \sqrt{ab}$.

Ans: Proof: see method.

5. Find the number of integers n with $1 \leq n \leq 1000$ such that $(n+3) \mid (n^2-1)$.

Ans: 2 $(n = 1$ and $n = 5)$